

MolecularEntity

A Schema.org Type

This term is in the "new" area - implementation feedback and adoption from applications and websites can help improve our definitions.

Thing > **BioChemEntity** > **MolecularEntity**

[\[more...\]](#)

Any constitutionally or isotopically distinct atom, molecule, ion, ion pair, radical, radical ion, complex, conformer etc., identifiable as a separately distinguishable entity.

Property	Expected Type	Description
Properties from MolecularEntity		
chemicalRole	DefinedTerm	A role played by the BioChemEntity within a chemical context.
inChI	Text	Non-proprietary identifier for molecular entity that can be used in printed and electronic data sources thus enabling easier linking of diverse data compilations.
inChIKey	Text	InChIKey is a hashed version of the full InChI (using the SHA-256 algorithm).

Property	Expected Type	Description
iupacName	Text	Systematic method of naming chemical compounds as recommended by the International Union of Pure and Applied Chemistry (IUPAC).
molecularFormula	Text	The empirical formula is the simplest whole number ratio of all the atoms in a molecule.
molecularWeight	QuantitativeValue or Text	This is the molecular weight of the entity being described, not of the parent. Units should be included in the form '<Number> <unit>', for example '12 amu' or as '<QuantitativeValue>'.
monoisotopicMolecularWeight	QuantitativeValue or Text	The monoisotopic mass is the sum of the masses of the atoms in a molecule using the unbound, ground-state, rest mass of the principal (most abundant) isotope for each element instead of the isotopic average mass. Please include the units in the form '<Number> <unit>', for example '770.230488 g/mol' or as '<QuantitativeValue>'.
potentialUse	DefinedTerm	Intended use of the BioChemEntity by humans.
smiles	Text	A specification in form of a line notation for describing the structure of chemical species using short ASCII strings. Double bond stereochemistry \ indicators may need to be escaped in the string in formats where the backslash is an escape character.
Properties from BioChemEntity		
associatedDisease	MedicalCondition or PropertyValue or URL	Disease associated to this BioChemEntity. Such disease can be a MedicalCondition or a URL. If you want to add an evidence supporting the association, please use PropertyValue.

Property	Expected Type	Description
bioChemInteraction	BioChemEntity	A BioChemEntity that is known to interact with this item.
bioChemSimilarity	BioChemEntity	A similar BioChemEntity, e.g., obtained by fingerprint similarity algorithms.
biologicalRole	DefinedTerm	A role played by the BioChemEntity within a biological context.
funding	Grant	A Grant that directly or indirectly provide funding or sponsorship for this item. See also ownershipFundingInfo . Inverse property: fundedItem
hasBioChemEntityPart	BioChemEntity	Indicates a BioChemEntity that (in some sense) has this BioChemEntity as a part. Inverse property: isPartOfBioChemEntity
hasMolecularFunction	DefinedTerm or PropertyValue or URL	Molecular function performed by this BioChemEntity; please use PropertyValue if you want to include any evidence.
hasRepresentation	PropertyValue or Text or URL	A common representation such as a protein sequence or chemical structure for this entity. For images use schema.org/image.
isEncodedByBioChemEntity	Gene	Another BioChemEntity encoding by this one. Inverse property: encodesBioChemEntity
isInvolvedInBiologicalProcess	DefinedTerm or PropertyValue or URL	Biological process this BioChemEntity is involved in; please use PropertyValue if you want to include any evidence.

Property	Expected Type	Description
isLocatedInSubcellularLocation	DefinedTerm or PropertyValue or URL	Subcellular location where this BioChemEntity is located; please use PropertyValue if you want to include any evidence.
isPartOfBioChemEntity	BioChemEntity	Indicates a BioChemEntity that is (in some sense) a part of this BioChemEntity. Inverse property: hasBioChemEntityPart
taxonomicRange	DefinedTerm or Taxon or Text or URL	The taxonomic grouping of the organism that expresses, encodes, or in some way related to the BioChemEntity.
Properties from Thing		
additionalType	Text or URL	An additional type for the item, typically used for adding more specific types from external vocabularies in microdata syntax. This is a relationship between something and a class that the thing is in. Typically the value is a URI-identified RDF class, and in this case corresponds to the use of rdf:type in RDF. Text values can be used sparingly, for cases where useful information can be added without their being an appropriate schema to reference. In the case of text values, the class label should follow the schema.org style guide .
alternateName	Text	An alias for the item.
description	Text or TextObject	A description of the item.
disambiguatingDescription	Text	A sub property of description. A short description of the item used to disambiguate from other, similar items. Information from other

Property	Expected Type	Description
		properties (in particular, name) may be necessary for the description to be useful for disambiguation.
identifier	PropertyValue or Text or URL	The identifier property represents any kind of identifier for any kind of Thing , such as ISBNs, GTIN codes, UUIDs etc. Schema.org provides dedicated properties for representing many of these, either as textual strings or as URL (URI) links. See background notes for more details.
image	ImageObject or URL	An image of the item. This can be a URL or a fully described ImageObject .
mainEntityOfPage	CreativeWork or URL	Indicates a page (or other CreativeWork) for which this thing is the main entity being described. See background notes for details. Inverse property: mainEntity
name	Text	The name of the item.
potentialAction	Action	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.
sameAs	URL	URL of a reference Web page that unambiguously indicates the item's identity. E.g. the URL of the item's Wikipedia page, Wikidata entry, or official website.
subjectOf	CreativeWork or Event	A CreativeWork or Event about this Thing. Inverse property: about
url	URL	URL of the item.

Source

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